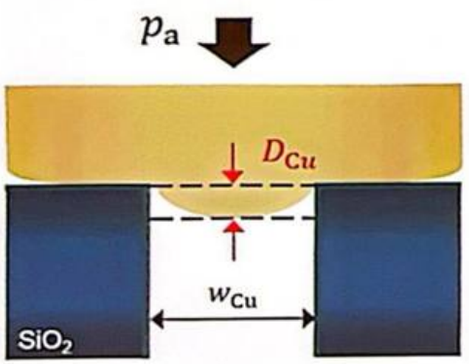
T27 CMP 접촉역학과 Cu dishing 제어

Sukkyung Kang · Hanyang University

Analytical model for dishing

$$\frac{\text{The pressure applied by an asperity}}{p_a} = \frac{\text{deformation pressure}}{p_{deform}} + \frac{\text{abrasion pressure}}{p_{abrasion}}$$



$$p_{deform} = \frac{E}{m(1-\nu^2)w_{Cu}} D_{Cu}$$

$$\frac{dD_{Cu}}{dt} = k_p p_{abrasion} v$$

$$D(t) = \left(\frac{m(1-\nu^2)w_{Cu}}{E}\right) p_a (1 - e^{-\alpha t})$$

Nomenclature

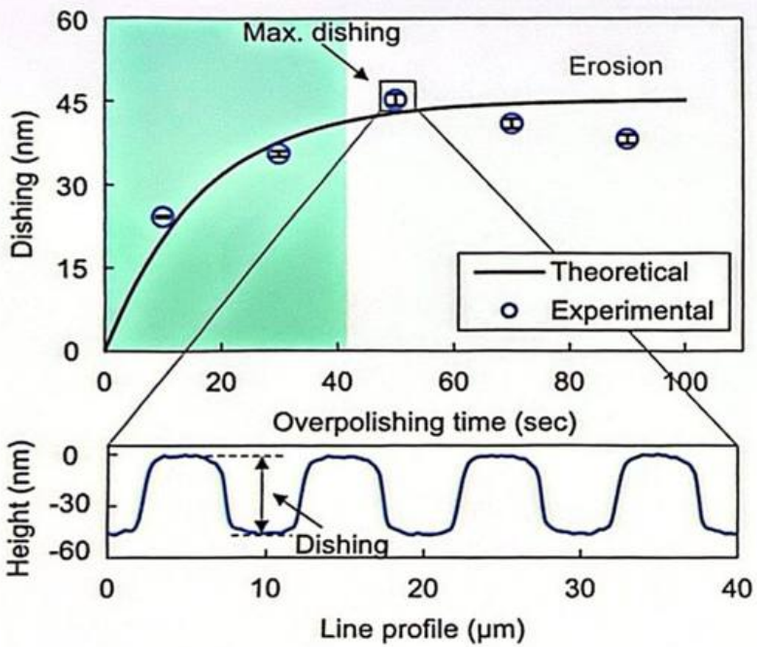
p_a: pressure applied by a single asperity
$p_{deforem}$: pressure consumed for pad deformation
$p_{abrasion}$: pressure available for material removal (abrasion)

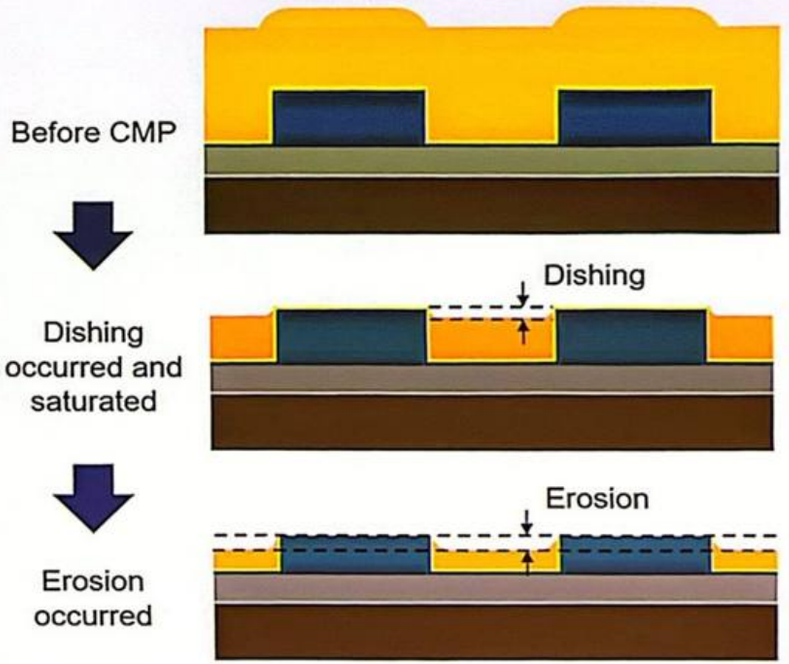
D_{Cu}: dishing depth of Cu
w_{Cu}: Cu trench width
k_p: Preston coefficient
v: relative polishing velocity
E: Young's modulus of pad asperity
ν: Poisson's Ratio of the pad asperity
m: fitting parameter
t: Overolishing time

V. H. Nguyen et al., Journal of the Electrochemical Society, 150, G689, 2003.

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Experimental results



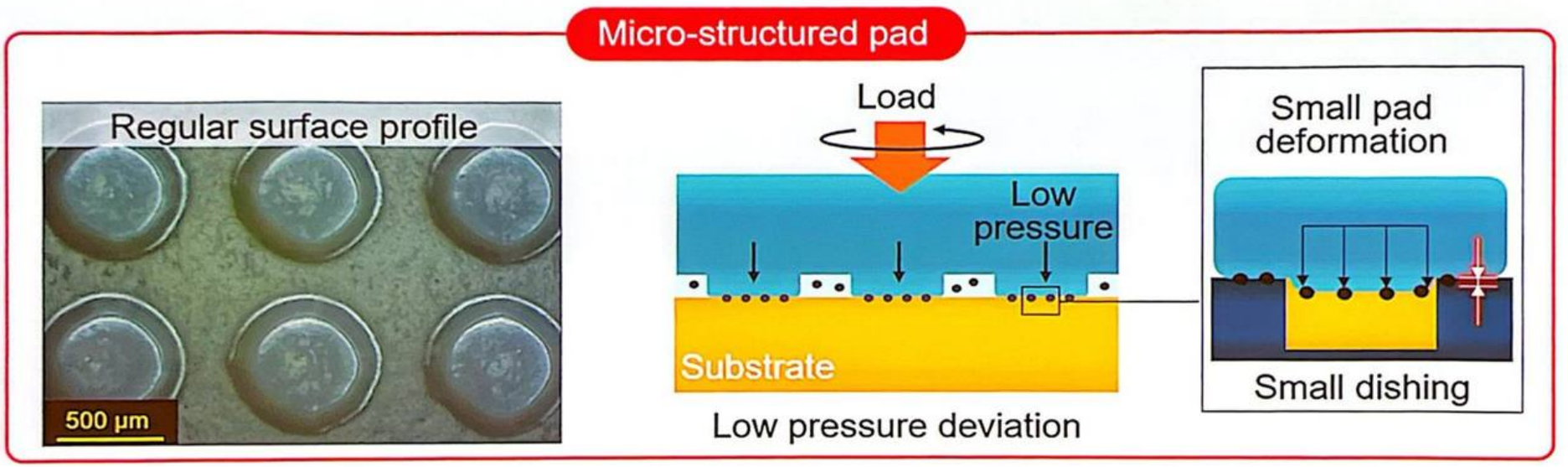


$$D_{\max} = \left(\frac{m(1-\nu^2)w_{\mathrm{Cu}}}{E}\right)p_{\mathrm{a}}$$

T27 CMP 접촉역학과 Cu dishing 제어

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Micro-structured CMP pad

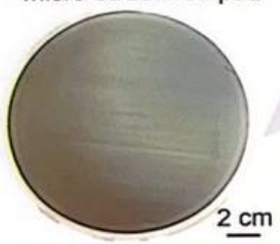


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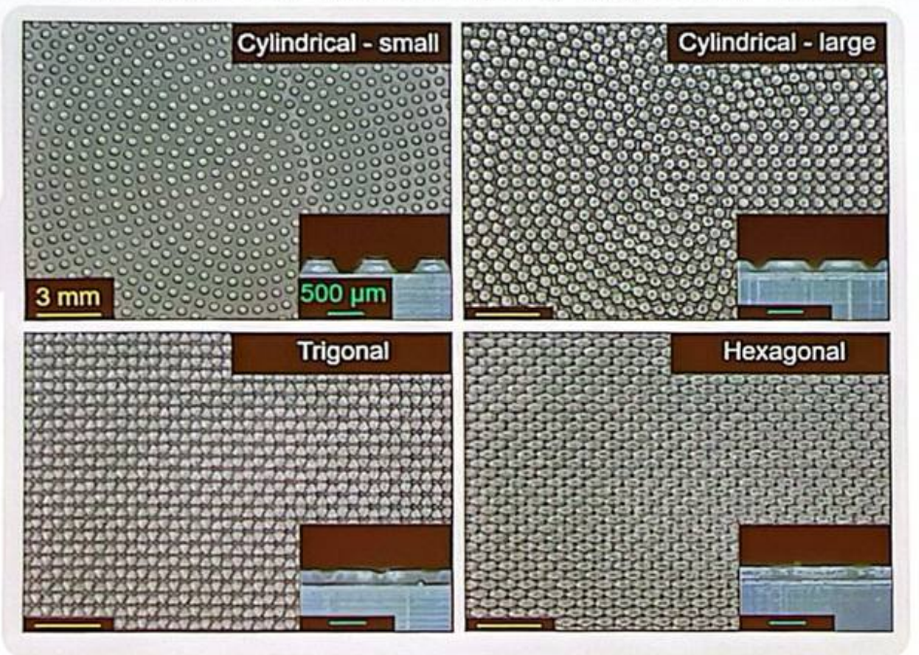
T27 CMP 접촉역학과 Cu dishing 제어

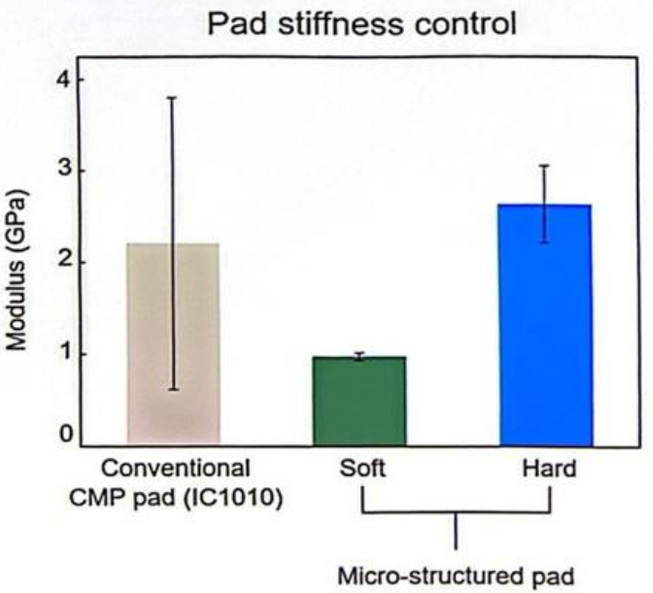
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Micro-structured CMP pad



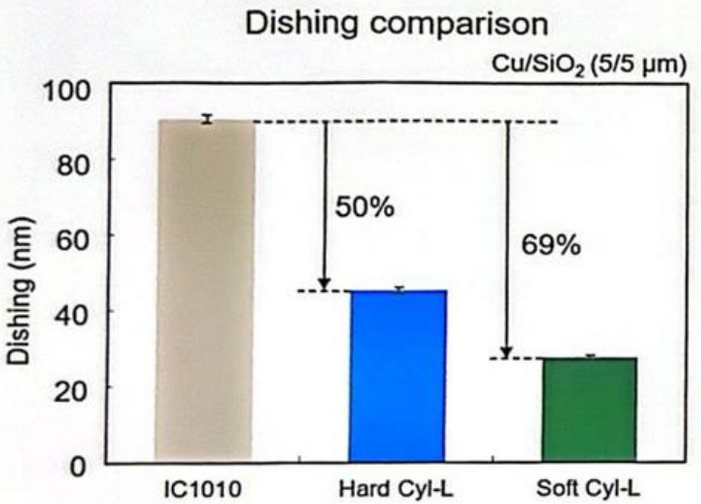
CMP pads with different asperity shapes and sizes

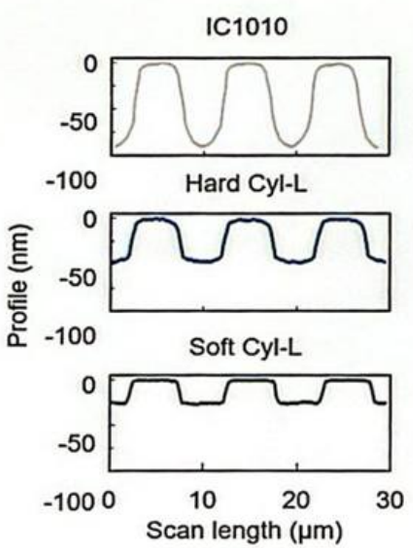




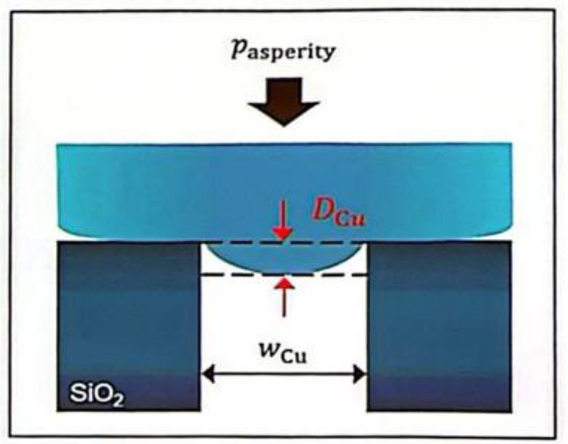
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Micro-structured pads reduced dishing





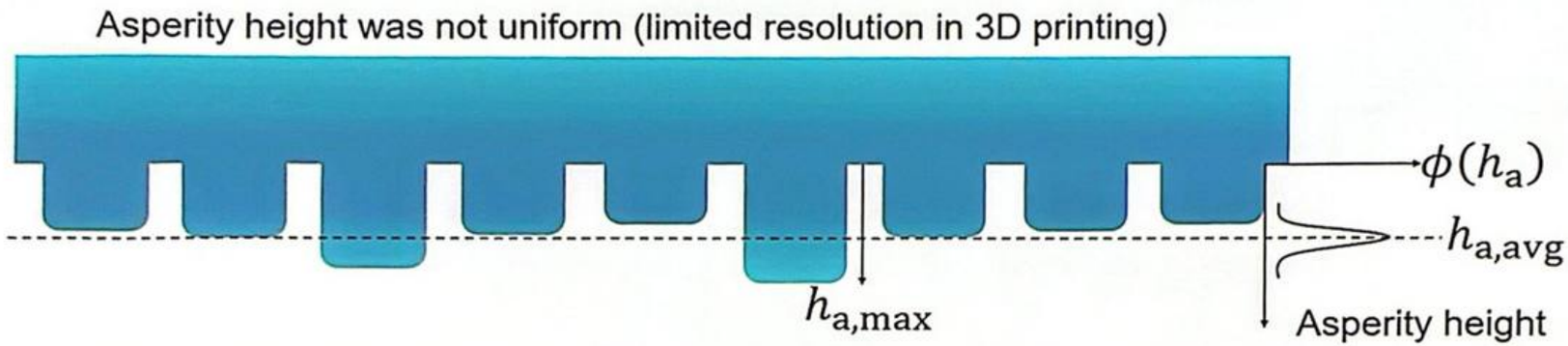
$$p_{asperity} = p_{deform} + p_{abrasion}$$



$$D_{max} = \left(\frac{m(1-\nu^2)w_{Cu}}{E}\right)p_a$$

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Asperity height variation



$$D_{max} = \left(\frac{m(1-\nu^2)w_{Cu}}{E}\right)p_{max}$$

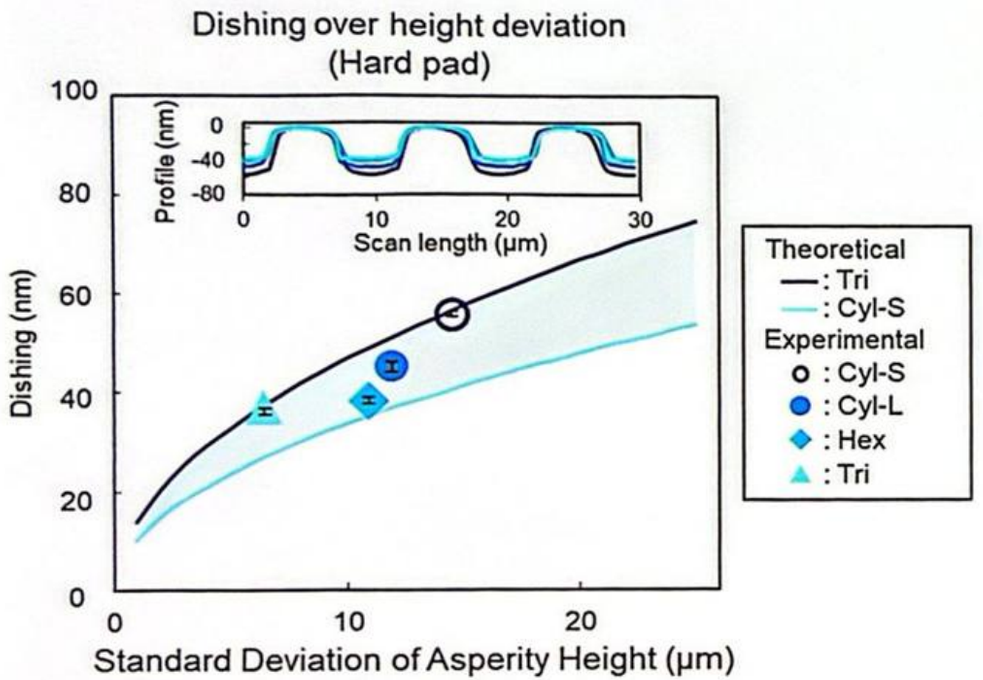
$$\Delta z_{max} = h_{a,max} - d$$

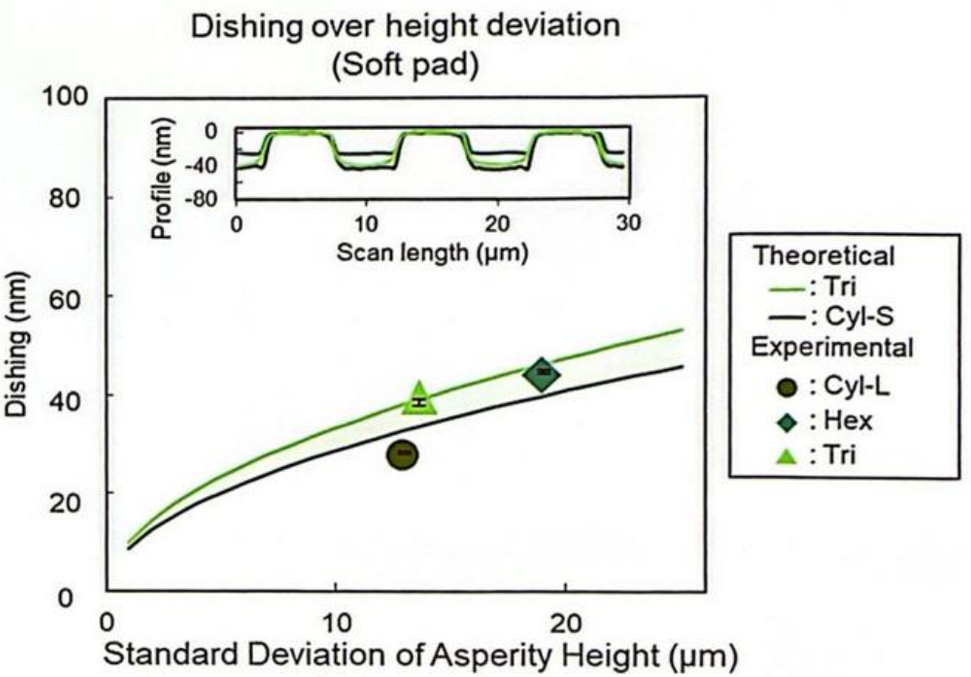
$$F_{max} = k\Delta z_{max}$$

$$p_{max} = \frac{k_a \Delta h_{a,max}}{A_a} = \frac{k_a(h_{a,max} - d)}{A_a}$$

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Incorporating asperity height variation in model





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Summary

- Controlling copper dishing is essential for hybrid bonding.
- We developed a dishing model that considers pad deformation and abrasion.
- We controlled the real contact pressure using micro-structured CMP pads.
- When asperity heights vary, maximum asperity local pressure is a key factor.

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